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ON THE NUCLEOLUS OF A CHARACTERISTIC FUNCTION GAME*

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Introduction. The nucleolus is a solution concept for games in characteristic function form. It was introduced by D. Schmeidler in [1]. This paper is concerned with some of the properties of the nucleolus. Some of the results are new (Theorems 2, 4, 5) whereas others are known (Theorems 1 and 3), but the proofs given here are, in a sense, simpler than those in [1].

A *characteristic function game* is a pair (N, v) consisting of a set $N = \{1, 2, \cdots, n\}$ of n *players*, and a *characteristic function* v, which maps each subset S of N, called a *coalition*, to a number $v(S)$. In addition, it is assumed that $v(N) \geqq 0$, and that $v(S) = 0$ for all one-person coalitions, as well as for the empty coalition. A *payoff vector* is an n-tuple $x = (x_1, \cdots, x_n)$ such that $x_i \geqq 0$ for all $i = 1, \cdots, n$, and $x(N) = v(N)$, where for each coalition S, $x(S)$ denotes $\sum_{i \in S} x_i$. X is the set of all payoff vectors. Of course, X depends on the particular game v.

For any x in X, let $q(x)$ be a vector in E^{2^n}, the components of which are the numbers $v(S) - x(S)$, arranged in order of decreasing magnitude, where S runs over all the coalitions of N. We shall say that x *is at least as acceptable as* y (with respect to v), and write $x \succsim y$, if $q(x)$ is not greater than $q(y)$, in the lexicographical order on E^{2^n}. If $q(x)$ is smaller than $q(y)$, we shall say that x *is more acceptable than* y and write $x \succ y$.

The *nucleolus* of the game is the set of those points in X which are "most acceptable," that is, $\{x \in X : x \succsim y \text{ for all } y \in X\}$.

In [1], Schmeidler proved the basic result that every game possesses a non-empty nucleolus. He further showed that the nucleolus actually consists of just one point, and that—viewed as a point function of the game—it is continuous. Our aim is to give new proofs of the uniqueness and of the continuity of the nucleolus. To this end, we will have to introduce a few definitions and to prove an auxiliary theorem (Theorem 2), which may have some interest in its own right.

DEFINITION. Let $b_0, b_1, \cdots, b_p$ be a sequence of sets whose elements are coalitions of N. This sequence is a *coalition array* whenever:

(i) every coalition of N is contained in exactly one of the sets $b_1, \cdots, b_p$,

(ii) b_0 contains only one-element coalitions.

For every game v and payoff vector x, let $b_1(x, v)$ be the set of those $S \subseteq N$ for which $\max\{v(S) - x(S) : S \subseteq N\}$ is attained. Similarly, $b_2(x, v)$ is the set of those $S \subseteq N$ where $\max\{v(S) - x(S) : S \notin b_1(x, v)\}$ is attained, and so on. Finally, let $b_0(x) = \{\{i\} : x_i = 0\}$. It is obvious that $b_0(x), b_1(x, v), \cdots, b_p(x, v)$ is a coalition array. We shall say that it is the array that *belongs* to (v, x).

DEFINITION 2. A coalition array $b_0, \cdots, b_p$ has *property* I if for all $k = 1, 2, \cdots, p$ and any y in E^n,

(1) $y(S) \geqq 0$ for all $S \in b_0$,

(2) $y(S) \geqq 0$ for all $S \in b_1 \cup \cdots \cup b_k$, and

(3) $y(N) = 0$

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imply

(4) $y(S) = 0$ for all $S \in b_1 \cup \cdots \cup b_k$.

LEMMA 1. *If x is the nucleolus of v, then the coalition array $b_0, \cdots, b_p$ that belongs to (v, x) has property* I.

Proof. Assume the contrary. Then a $y \in E^n$ exists such that for some k, y satisfies (1), (2) and (3) of Definition 2, and for at least one S in $b_1 \cup \cdots \cup b_k$, $y(S) > 0$.

Condition (1) implies that $y_i \geqq 0$ whenever $x_i = 0$. It follows that $x + ty \geqq 0$ for small positive t. Since $y(N) = 0$, $x + ty$ is in X for small positive t.

Let $v(S) - x(S) = a_1$ for S in b_1, $v(S) - x(S) = a_2$ for S in b_2, etc. Let us further denote the number of coalitions appearing in $b_1, \cdots, b_k$ by $m_1, \cdots, m_k$, respectively. Then the first m_1 components of $q(x)$ are a_1, the next m_2 components are a_2, etc. Now if we choose our positive t to be small enough, then the first m_1 components of $q(x + ty)$ will be the numbers $v(S) - x(S) - ty(S)$, $S \in b_1$, the next m_2 components will be the numbers $v(S) - x(S) - ty(S)$, $S \in b_2$, etc. Since $y(S) \geqq 0$ for all $S \in b_1 \cup \cdots \cup b_k$, none of the first m_1 components of $q(x + ty)$ is greater than a_1, none of the next m_2 components is greater than a_2, etc. So we have that all of the first $m_1 + \cdots + m_k$ components of $q(x + ty)$ are not larger than the corresponding components of $q(x)$. Now since $y(S) > 0$ for at least one S in $b_1 \cup \cdots \cup b_k$, it follows that at least one of these $m_1 + \cdots + m_k$ components is smaller than its corresponding component. Thus $q(x + ty)$ is smaller in the lexicographical order than $q(x)$, and $x + ty$ is more acceptable than x. This contradicts the fact that x is in the nucleolus, and the proof is complete.

LEMMA 2. *If the condition array that belongs to (v, x) possesses property* I, *then x is more acceptable (with respect to v) than any y in X which differs from x.*

Proof. Assume that $y \in X$ is at least as acceptable as x. Let $b_0, \cdots, b_p$ be the coalition array that belongs to (v, x). Let $m_1, \cdots, m_p$ denote the number of coalitions which belong to $b_1, \cdots, b_p$. Consider $q(x)$. Its first m_1 components have magnitude $v(S) - x(S)$, where S is any coalition of b_1. Its next m_2 components have magnitude $v(S) - x(S)$, where S is any coalition in b_2, and so on. Since $y \succsim x$, $v(S) - y(S)$, for any S, must be no greater than the first component of $q(x)$. In particular, $v(S) - y(S) \leqq v(S) - x(S)$ for $S \in b_1$, so that

$$(y - x)(S) \geqq 0, \qquad S \in b_1. \tag{2a}$$

Also, since y is nonnegative and x vanishes on b_0, we have

$$(y - x)(S) \geqq 0, \qquad S \in b_0. \tag{1a}$$

Finally, since $x(N) = v(N) = y(N)$,

$$(y - x)(N) = 0. \tag{3a}$$

(1a), (2a) and (3a), together with property I (for $k = 1$), imply $(y - x)(S) = 0$ for all S in b_1.

But this shows that $q(y)$ contains m_1 components which are as large as the first m_1 components of $q(x)$. Thus $y \succsim x$ now implies that all other components of $q(y)$ must be no greater than the $(m + 1)$st component of $q(x)$. That is, $v(S) - y(S)$, for any $S \notin b_1$, is no greater than $v(S) - x(S)$, $S \in b_2$. In particular, $(y - x)(S) \geqq 0$

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for S in b_2. So now we have

$$(1b) \qquad (y - x)(S) \geqq 0, \qquad S \in b_0,$$

$$(2b) \qquad (y - x)(S) \geqq 0, \qquad S \in b_1 \cup b_2,$$

$$(3b) \qquad (y - x)(N) = 0.$$

And so, by property I (for $k = 2$),

$$(y - x)(S) = 0, \qquad S \in b_1 \cup b_2.$$

Continuing along the same line of reasoning, we arrive at the conclusion $(y - x)(S) = 0$ for all $S \in b_1 \cup \cdots \cup b_p$. Hence $y = x$, and the proof is complete.

A consequence of Lemmas 1 and 2 is that if x is at least as acceptable as all other points in X, then it is more acceptable than all other points in X.

THEOREM 1 (Schmeidler). *The nucleolus consists of a unique point.*

Also, we have the following theorem.

THEOREM 2. *A payoff vector x is the nucleolus of v if and only if the coalition array that belongs to (v, x) has property* I.

Let us consider an arbitrary but fixed order $S_1, S_2, \cdots, S_{2^n}$ on the coalitions of N. Every game can be uniquely represented as a vector in E^{2^n} by setting $v_i = v(S_i)$. Let V be the set of all n-person characteristic functions considered as a subset of E^{2^n}. Let $f(v)$ denote the nucleolus of v. f is thus a function from a subset of E^{2^n} to a subset of E^n.

Before we go on to the next result, let us introduce the following definition.

DEFINITION 3. The coalition array $c_0, \cdots, c_r$ is *derived* from the coalition array $b_0, \cdots, b_p$ if $c_0 \supseteq b_0$, $c_1 = b_1 \cup \cdots \cup b_{k_1}$, $c_2 = b_{k_1+1} \cup \cdots \cup b_{k_2}$, etc. for some $1 \leqq k_1 < k_2 < \cdots < k_r = p$.

It is easy to see that if an array has property I, then any array which is derived from it also has property I.

THEOREM 3 (Schmeidler). *$f: V \to E^n$ is a continuous function.*

Proof. Let V_1 denote the subset of V consisting of those characteristic functions v with $v(N) = 1$, and let f_1 be the restriction of f to V_1. Now $f(v) = v(N)f_1(v/v(N))$. Hence to prove the theorem, it is sufficient to prove that f_1 is continuous. All v in V_1 have the same set X of payoff vectors, and this set is compact. It therefore suffices to prove that the graph $\{(v, x): v \in V_1, x \text{ is the nucleolus of } v\}$ is closed. By Theorem 2, this is the same as showing that the set $\{(v, x): v \in V_1$, the array that belongs to (v, x) has property I$\}$ is closed. So, let $v^n \to v$, $x^n \to x$, where each (v^n, x^n) has property I. We must show that (v, x) also has property I. As the number of coalition arrays is finite, we may assume, without loss of generality, that all the (v^n, x^n) have the same array: $b_0, \cdots, b_p$. This array obviously has property I. Since all weak inequalities are preserved in passing to the limit, it follows that the array that belongs to (v, x) is derived from $b_0, \cdots, b_p$, and so it has property I.

Before we go on, let us write down another result, which is a straightforward application of Theorem 2 and Theorem 3.

THEOREM 4. *Let $b_0, \cdots, b_p$ be a coalition array that has property* I. *Let $A = \{v \in V$: the coalition array that belongs to $(v, f(v))$ is derived from $b_0, \cdots, b_p\}$.*

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Then A is a closed and convex cone, and moreover, if v_1, v_2 are in A, then

$$f(t_1v_1 + t_2v_2) = t_1f(v_1) + t_2f(v_2) \qquad \textit{for all } t_1, t_2 \geqq 0.$$

Proof. A is closed because of the continuity of f.

Now, if $v_1, v_2 \in A$, it is easy to see that the coalition array that belongs to $(t_1v_1 + t_2v_2, t_1f(v_1) + t_2f(v_2))$ is derived from $b_0, \cdots, b_p$, which means that it has property I, and so $t_1f(v_1) + t_2f(v_2)$ is the nucleolus of $t_1v_1 + t_2v_2$.

Theorem 4 means that the set V of characteristic function games is a union of a finite number of closed and convex cones, on each of which the nucleolus is a linear function of v (cf. [4], [5]).

To conclude, we would like to show that property I may be given an equivalent formulation, using the notion of a "balanced" collection of coalitions. We denote by (x, y) the scalar product of x and y.

LEMMA 3. *Let $a_1, \cdots, a_m$ and b be nonnegative vectors in E^n. Let c be a nonnegative, nonzero vector in E^n. Suppose that for each y in E^n,*

$$(a_i, y) \geqq 0, \quad i = 1, \cdots, m, \quad \textit{and} \quad (b, y) = 0 \quad \textit{imply} \quad (c, y) \leqq 0.$$

Then,

$$b = \sum_{i=1}^{m} t_ia_i + tc, \qquad \textit{where } t_i \geqq 0 \ \textit{ and } \ t > 0.$$

Proof. We have:

$$\left.\begin{array}{ll} (a_i, y) \geqq 0, & i = 1, \cdots, m, \\ (b, y) \geqq 0, & \\ (-b, y) \geqq 0 & \end{array}\right\} \Rightarrow (-c, y) \geqq 0.$$

So, by the well-known theorem of Farkas [2], we have

$$-c = \sum_{i=1}^{m} q_ia_i + qb, \qquad \text{where } q_i \geqq 0.$$

The nonnegativity of the vectors implies $q < 0$, and a division by q yields the required result.

LEMMA 4. *Let $a_1, \cdots, a_m$ and b be nonnegative vectors in E^n. Let $c_1, \cdots, c_k$ be nonnegative, nonzero vectors in E^n. Suppose that for each y in E^n,*

$$\left.\begin{array}{ll} (a_i, y) \geqq 0, & i = 1, \cdots, m, \\ (b, y) = 0, & \\ (c_j, y) \geqq 0, & j = 1, \cdots, k, \end{array}\right\} \Rightarrow (c_j, y) = 0, \qquad j = 1, \cdots, k.$$

Then,

$$b = \sum_{i=1}^{m} u_ia_i + \sum_{j=1}^{k} w_jc_j, \qquad \textit{where } u_i \geqq 0 \textit{ and } w_j > 0.$$

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Proof. For each $j_0 = 1, \cdots, k$, we have

$$\left.\begin{array}{ll} (a_i, y) \geqq 0, & i = 1, \cdots, m, \\ (c_j, y) \geqq 0, & j = 1, \cdots, k, \\ (b, y) = 0 & \end{array}\right\} \Rightarrow (c_{j0}, y) \leqq 0.$$

Hence, by Lemma 3,

$$b = \sum_{i=1}^{m} r_i a_i + \sum_{j=1}^{k} s_j c_j + t c_{j0},$$

where $r_i, s_j \geqq 0$ and $t > 0$.

Adding all these equations for $j_0 = 1, \cdots, k$, and then dividing by k, yields the required result.

Let e_S denote the characteristic vector of the coalition S; that is, $(e_S)_i = 1$ for $i \in S$, and $(e_S)_i = 0$ for $i \notin S$.

DEFINITION 4. A coalition array $b_0, \cdots, b_p$ has *property* II if for each $k = 1, \cdots, p$, there is a 2^n-dimensional vector $w = w(k)$, $w \geqq 0$, such that

$$\sum_{S \in b_0 \cup \cdots \cup b_k} w_S e_S = e_N \quad \text{and} \quad w_S > 0 \quad \text{for } S \in b_1 \cup \cdots \cup b_k.$$

Remark. Property II is equivalent to the following condition: For each $k = 1, \cdots, p$, there is a subset b_0^k of b_0 such that $b_0^k \cup b_1 \cup \cdots \cup b_k$ is balanced (for the definition of a "balanced" collection of sets see [3]).

THEOREM 5. *A payoff vector x is the nucleolus of v if and only if the coalition array that belongs to (v, x) has property* II.

Proof. By Theorem 3, it is sufficient to show that properties I and II are equivalent. One may readily verify, by direct calculation, that property II implies property I. The converse implication follows from Lemma 4, once we note that $(e_S, y) = y(S)$.

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